DERIVED GLOBAL SECTIONS AGAINST A PERFECT OBJECT ARE PSEUDO-COHERENT AND COMMUTE WITH BASE CHANGE ALONG THE DERIVED DESCENT TOWER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a stage-wise RGamma computation with base change compatible pseudo-coherent transitions, the derived-limit commutation of RGamma(X, -), and the algebra limit lemma for pseudo-coherent objects

Editorial note. The label lemma-compute-against-perfect-derived names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The derived descent tower situation with its system of pseudo-coherent objects K_n and the limit object K defined as the derived limit in the quasi-coherent derived category
- The derived categories of spaces lemma that the direct image computing the stages of a pseudo-coherent system is pseudo-coherent with formation commuting with base change
- The More on Algebra lemma that a derived limit of pseudo-coherent objects with compatible base change data is pseudo-coherent and recovers the stages on base change
- The exact functor RGamma on the quasi-coherent derived category commuting with products and hence with derived limits

Source reference. Original reference 0DIC: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2695–2695. Prerequisite pointer invoked at proof line(s) 2755; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CQF: sources/stacks-chow/tags.txt, source lines 15960–15960. Prerequisite pointer invoked at proof line(s) 2777; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DIB: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2668–2668. Prerequisite pointer invoked at proof line(s) 2754; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0CTN: sources/stacks-chow/tags.txt, source lines 16073–16073. Prerequisite pointer invoked at proof line(s) 2771; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0DIB let K be as in Lemma 0DIC. For any perfect object E of $D(\mathcal{O}_X)$ the cohomology*

$$M = R\Gamma(X, K \otimes^{\mathbf{L}} E)$$

is a pseudo-coherent object of $D(A)$ and there is a canonical isomorphism

$$R\Gamma(X_n, K_n \otimes^{\mathbf{L}} E|_{X_n}) = M \otimes^{\mathbf{L}}_A A_n$$

in $D(A_n)$. Here $E|_{X_n}$ denotes the derived pullback of E to X_n.

Proof. Write $E_n = E|_{X_n}$ and $M_n = R\Gamma(X_n, K_n \otimes^{\mathbf{L}} E|_{X_n})$. By Derived Categories of Spaces, Lemma 0CTN we see that M_n is a pseudo-coherent object of $D(A_n)$ whose formation commutes with base change. Thus the maps $M_n \otimes^{\mathbf{L}}_{A_n} A_{n-1} \to M_{n-1}$ coming from φ_n are isomorphisms. By More on Algebra, Lemma 0CQF we find that $R\lim M_n$ is pseudo-coherent and that its base change back to A_n recovers M_n. On the other hand, the exact functor $R\Gamma(X, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ of triangulated categories commutes with products and hence with derived limits, whence

$$R\Gamma(X, E \otimes^{\mathbf{L}} K) = R\lim R\Gamma(X, E \otimes^{\mathbf{L}} K_n) = R\lim R\Gamma(X_n, E_n \otimes^{\mathbf{L}} K_n) = R\lim M_n$$

as desired. □